

1. Explain the difference between a “Fixed Effect” and a “Random Effect” in a linear model.

Fixed effect models assume that each parameter has one true value, its variability being due to sampling error alone. That is, in a linear model, a fixed parameter would have the same value for each observation in the dataset. Random effects models assume that some (as in mixed effect models) or all of the parameters follow some distribution, often a realization from a normal distribution, thus not having one true value. So, a random effects parameter can have different values depending upon the observation/individual/subject in the data.

2. List the three types of “Mixed Effects” models described in the text and provide a brief description of each type and examples of when they might be used.

- Random Effects

- Description: These models assume a parameter to have a potentially different value for each subject/observation in the dataset. That is, the parameter is a random variable sampled from some normal distribution, $p_i \sim N(0, \sigma_p^2)$. The model becomes a mixed effects model when these random variable parameters are included with fixed effect parameters.
- Examples:
 - We want to determine the effect of a drug on blood pressure, but believe that there are some fundamental differences between patients which have an effect on the target variable. If we assume the effect of each patient is random and *not* constant, we can implement a parameter as a random effect. It follows that the parameter would have its own variation, increasing the overall variability of the model but potentially fitting the data more adequately. Motivating factors behind using the parameter as a random, non-fixed effect are if there is high variability in patient effects, and if there is a need to adjust for this effect.
 - An example in the psychology field could be a study where researchers are interested in determining the effectiveness of cognitive behavioral therapy (CBT) on reducing recidivism. It seems logical that not every individual in the study is fundamentally the same, and assuming these differences are not systematic but random, it follows that a *random* effects parameter may be useful in capturing these differences.

- Random Coefficient

- Description: Random coefficient models treat the coefficients of the parameters as random variables. That is, for each subject in the dataset, the model may fit a different value for slope and intercept. In other words, each observation will have

its own regression line. This could be notated by β_{0i} , representing the intercept of the i th observation, and β_{1i} , representing the slope of the i th observation. In this type of mixed model the coefficients themselves are random and not necessarily the same for each subject, hence the name.

- Examples:

- If we are studying the weight of rats based on particular foods over time, it may be logical to account for the fact that not all of the rats will begin the study at the same weight. Additionally, due to biological (metabolical) differences, they may gain or lose weight at different rates or in a different manner. These two facets may make the inclusion of subject-specific intercepts and slopes, respectively, useful. If two of the rats differ significantly in weight at the beginning of the study, a single regression line would be expected to have a large error. Thus accounting for this discrepancy with random coefficients may improve the accuracy and utility of the model.
- The performance of students on exams in different disciplines, such as mathematics, chemistry, history, etc over time. The expected performance on these may differ greatly due to the differences between each field of study, so it may be useful to fit a model that has a different regression line for each field.

- Covariance Pattern

- Description: These models are primarily motivated by the existence of serial correlation in the dataset. That is, we model the within-subject covariance matrix to account for the relationship between a variable and itself at different time points. The serial correlation would be removed to avoid any leftover pattern in the residuals. This can be done in three ways: 1 - if the serial correlation is constant regardless of the lag, 2 - the correlation decays as the amount of lag increases, 3 - each pair of observations has its own correlation.

- Examples:

- This model would be most common in time-series datasets, in which multiple measurements are taken on the same variable over a period of time. We can re-use the rats example here, since it is an example of a time series dataset. Intuition suggests that the weight of a rat at the current week is highly correlated to its weight from last week. It then follows that we would want to capture this serial correlation. This situation would likely be accounted for in the first or second way as described above, although all three ought to be checked.
- A model predicting the performance of spacecraft maneuvers may also benefit from a covariance pattern mixed effect model. Data observed from one time point could be highly correlated with data from another time

point, which is intuitive because it is being measured on the same craft. Capturing this relationship may be useful in simulating any jumps or outliers observed in the raw data, as the extra source of variability is being accounted for.

3. FEV1 data were collected on 10 subjects, five in each of two different treatment groups A and B, at baseline and every 6 months for a two year period (i.e. up to 5 observations per subject). Since there was missing data and not all observations were collected at exactly the same time intervals the data were fit using a “Mixed Effect” model.

The form of the model is given by:

$$FEV_{1ik} = \hat{\alpha}_0 + \hat{\alpha}_1 Age + \hat{\alpha}_2 BFEV_1 + \widehat{\beta_{0i}} + \widehat{\beta_{1i}} Age$$

$i = 1, 2, \dots, 10$ (subjects), $k = 1, 2, \dots, 4$ (observations)

$BFEV_1$ = Baseline FEV1, Alphas are Fixed Effects, betas are Random Effects

a) What type of mixed effect model is this?

This is a random coefficients model, where $\widehat{\beta_{0i}}$ represents the difference in intercept for the i th subject, and $\widehat{\beta_{1i}}$ represents the difference in slope for the i th subject. Both parameters are random, realizations from a normal distribution.

b) Why is it good practice to include the baseline FEV1 in this model? Give an example how it would adjust for subjects with high and low initial values differently.

A parameter for baseline FEV1 is useful because each patient likely starts at a different value. It is important to capture this fact because differing initial values can lead to very different values later on, unique to each patient. The parameter is fixed, meaning it does not change for any of the subjects. If a subject started with an abnormally high FEV1, it is expected for the ending FEV1 to be higher. On the other hand, if a subject started with a low initial value, they would likely have a lower FEV1. The parameter α_2 essentially gives a weight to this initial value, that is a higher absolute value of α_2 suggests the baseline FEV1 has a very strong effect on the final FEV1. Since we are assuming a linear relationship between the initial FEV1 and the ending FEV1, it is useful to include a parameter that measures this relationship directly.

c) If we assume that the random variables are from a bivariate normal distribution where;

$$\begin{pmatrix} \widehat{\beta}_{0i} \\ \widehat{\beta}_{1i} \end{pmatrix} \sim N(0, G), \text{ with } G = \begin{bmatrix} 0.73 & 0.13 \\ 0.13 & 1.21 \end{bmatrix}$$

Give an interpretation of each of the elements in this covariance matrix.

The variance of $\widehat{\beta}_{0i}$ is 0.73, the between-treatments effect, while the variance of $\widehat{\beta}_{1i}$ is 1.21. The latter therefore has greater variability. The remaining elements, each 0.13, are the covariance between $\widehat{\beta}_{0i}$ and $\widehat{\beta}_{1i}$, meaning they have a very slight positive relationship (Corr ≈ 0.1383). These elements have the same value because Cov(A,B) is equivalent to Cov(B,A). Depending on the scale of the data, any of the aforementioned values could indicate large variability or little variability.

d) Using this same matrix G, if the covariance term is found to not be statistically significantly different from zero, what does that tell us about the relation between our random variables?

This tells us that the random variables do not have a significant effect on one another. That is, if one increases, there is no statistical evidence suggesting that the other will increase OR that it will decrease. They would be statistically unrelated, or independent (which follows because both parameters are normally distributed).

4. Explain the difference between a Random Effects Model and Random Coefficient Model and discuss any assumptions and differences in assumptions between the two.

A random effects model and random coefficient model differ in the implication of their parameters. While both have parameters with unique values for specific subjects in the dataset, a random effects parameter captures the effect of the indicated subject directly, whereas a random coefficient parameter essentially changes the entire model according to the specific subject. Another way to describe it is that a random effects parameter has a different value for any i th subject, changing the intercept, while a random coefficients parameter changes the slope. Random effects assume that the residuals are normally distributed with 0 mean and constant variance σ^2 , with each random effect parameter also univariate-normally distributed with 0 mean and its own constant variance σ_p^2 . It is also assumed that the random effects parameters are independent of each other, that is there is no correlation between them.

Random coefficients models have the same assumptions on the residual terms, but differs slightly in that its slope and intercept parameters follow a bivariate normal distribution, its variance the covariance matrix between the parameters as n slopes and n intercepts will be fitted to the model. This leads to an extra source of variability and dependency structure within the

model that ought to be accounted for, and often it is captured by the use of covariance pattern models.

5. *A Random Coefficient Model was fit to longitudinal data from a study looking at reductions in cardiac output as related to a new anesthetic. Measures were taken every 30 minutes for the course of the operation on 20 patients. Patients and patient by time interactions were fitted as random effects. The following covariance matrix was obtained for the random coefficients,*

$$\begin{matrix} sp^2 & sp*pt \\ sp*pt & spt^2 \end{matrix}$$

$$G = \begin{matrix} sp^2 & sp*pt \\ sp*pt & spt^2 \end{matrix}$$

$$sp^2 = 1.23, sp*pt = 0.513, spt^2 = 0.783$$

a. *Explain what the variance terms in the model are assessing.*

sp^2 assesses the variance of the random intercept parameter. That is, how much variation is in the set of all intercepts in the model - one for each observation. spt^2 assesses the variance of the random slope parameter, or how much variation there is among all of the unique slope values. If these values are not significantly different to 0, it follows that there may not be a need for a random coefficient component because the regression line does not vary significantly from subject to subject. This effectively means that the model is capturing variance in the data where there is not significant variance to capture, overcomplicating and possibly overfitting the model. The off-diagonal terms $sp \cdot pt$ assess the covariance (linear relationship) between the slope and intercept parameters. A larger value indicates a stronger relationship between the two, although this should be scaled by the variance of each parameter to calculate their correlation coefficient. If this correlation coefficient is close to 1, it means that the model fits positively-sloped lines that grow in steepness as the intercept rises up the y-axis. A correlation coefficient close to 0 means that a particular intercept is not expected to have any particular slope - it could be positive, negative, close to 0. Correlation close to -1 suggests that the model fits lines of increasing steepness as the intercept rises but in the negative direction. A way to visualize this would be to imagine the lines converging onto the y-axis as the intercept rises, as if you're pulling a string with a ball attached to the end of it up a wall.

b. *Give an interpretation of the covariance term in this model. What does it mean if these terms are not statistically significant? Will the two covariance terms always be equal? Why or why not?*

A positive covariance term here means that the slopes increase as the intercept increases. That is, the slopes and intercepts are positively correlated. If the covariance terms are not statistically significant, it means the random coefficients do not have a significant effect on another. So if either increases or decreases, it gives no indication on whether the other will increase or decrease. These terms will always be equal, because $Cov(A, B) = Cov(B, A)$, and the covariance matrix is a symmetric matrix.

6. The following data were obtained in a longitudinal study comparing levels of FEV1 between treatments A and B on four different subjects,

Observation	Subject	FEV1	Time	Treatment
1	1	3.56	0	A
2	1	3.90	5.6	A
3	1	4.21	7.5	A
4	2	5.12	0	B
5	2	5.55	6.6	B
6	3	4.49	0	A
7	3	4.66	4.9	A
8	3	4.71	7.2	A
9	4	2.99	0	B
10	4	3.25	3.6	B

Note! (For all sections below show the “over-parameterized” design matrices)

a) Using these data show what the design matrices X and parameter vector A must be to fit a general linear model predicting FEV1 as a function of time and treatment.

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 5.6 & 1 & 0 \\ 1 & 7.5 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 6.6 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 4.9 & 1 & 0 \\ 1 & 7.2 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 3.6 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix},$$

where μ represents the overall mean, α_1 the time value for the patient, α_2 a dummy variable where 1=treatment A, and α_3 a second dummy variable where 1=treatment B.

b) Show how the design matrix must change if the model were to include a time squared term.

$$X = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 5.6 & 31.36 & 1 & 0 \\ 1 & 7.5 & 56.25 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 6.6 & 43.56 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 4.9 & 24.01 & 1 & 0 \\ 1 & 7.2 & 51.84 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 3.6 & 12.96 & 0 & 1 \end{bmatrix}$$

This model now has 5 parameters, with α_1 corresponding time, α_2 time squared, with α_3 and α_4 the dummy variables indicating which treatment was administered. Overall mean μ remains.

c) Using these data show what the design matrices X and Z must be to fit a random effects model predicting FEV1 as a function of time and treatment, and treating subjects as a random effect.

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 5.6 & 1 & 0 \\ 1 & 7.5 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 6.6 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 4.9 & 1 & 0 \\ 1 & 7.2 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 3.6 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix}, \text{ where } \beta_i \text{ represents the effect of the } i\text{th subject on FEV1}$$

d) Using these data show what the design matrices X and Z must be to fit a random coefficient model predicting FEV1 as a function of time and treatment, and treating subjects and subjects by time as a random effect.

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 5.6 & 1 & 0 \\ 1 & 7.5 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 6.6 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 4.9 & 1 & 0 \\ 1 & 7.2 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 3.6 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix},$$